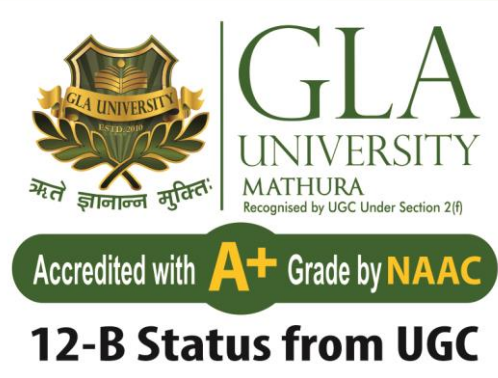


Microprocessor 8085

PIN Diagram Description



By

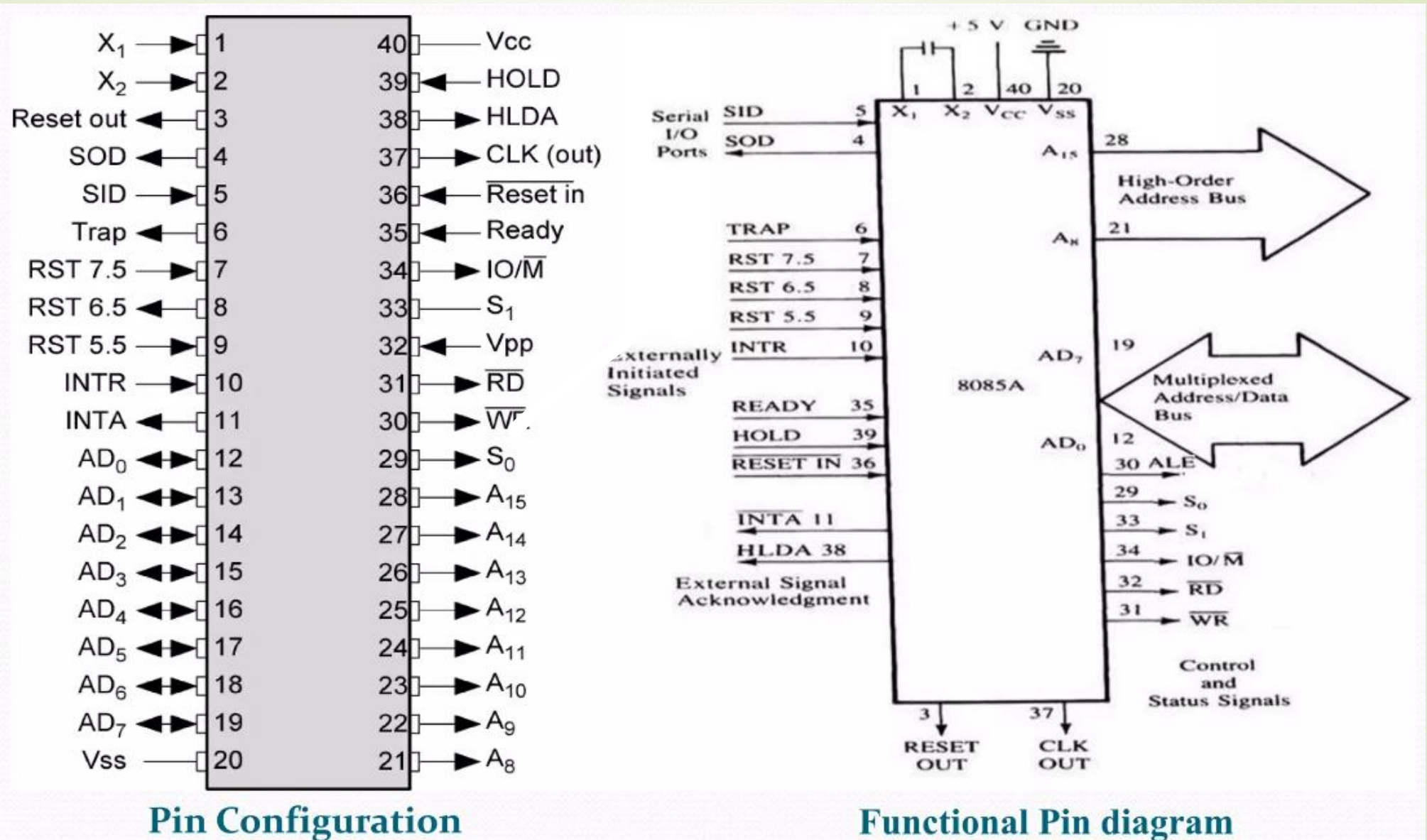
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Outline

- PIN Diagram of 8085
- Specifications
- Crystal Oscillator-Power Supply-Clock
- Buses, Control Signals and Status Configuration
- DMA and Interrupt Signals

Pin Description in functional and configuration form



Specifications

- The 8085 is an 8-bit general purpose microprocessor that can address 64K Byte of memory.
- It has 40 pins and uses +5V for power. It can run at a maximum frequency of 3 MHz.
- The pins on the chip can be grouped into 6 groups:
 - Address Bus and Data Bus.
 - Status Signals.
 - Control signal
 - Interrupt signal
 - Power supply and Clock signal
 - Reset Signal
 - DMA request Signal
 - Serial I/O signal
 - Externally Initiated Signals.

Specifications Cont....

8085 is developed by INTEL

8 bit microprocessor: can accept 8 bit data simultaneously

Operates on single +5V D.C. supply.

Designed using NMOS technology

6200 transistor on single chip

It provides on chip clock generator, hence it does not require external clock generator.

Operates on 3MHz clock frequency.

8bit multiplexed address/data bus, which reduce the number of pins.

16address lines, hence it can address $2^{16} = 64$ K bytes of memory

It generates 8 bit I/O addresses, hence it can access $2^8 = 256$ I/O ports.

5 hardware interrupts i.e. TRAP, RST6.5, RST5.5, RST4.5, and INTR

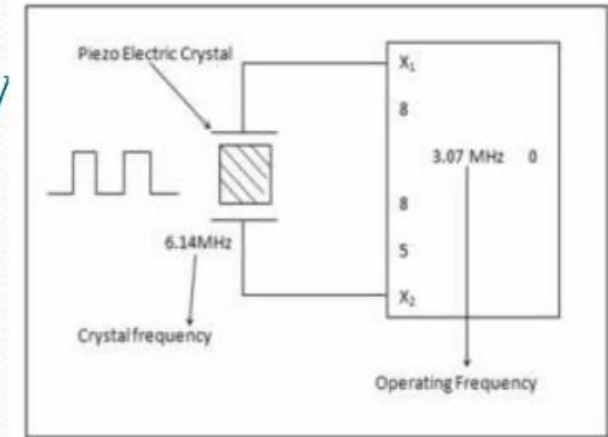
It provides DMA.

Crystal Oscillator, Vcc, Vss and Clock

Vcc (**Pin 40**) : single +5 volt power supply

Vss (**Pin 20**) : Ground

There are 3 important pins in this group.



✓ Xo and X1 :((**Pin 1-2**)

- Crystal or R/C network or LC network connections to set the frequency of internal clock generator.
- The frequency is internally divided by two.
- Since the basic operating timing frequency is 3 MHz, a 6 MHz crystal is connected to the Xo and X1 pins.

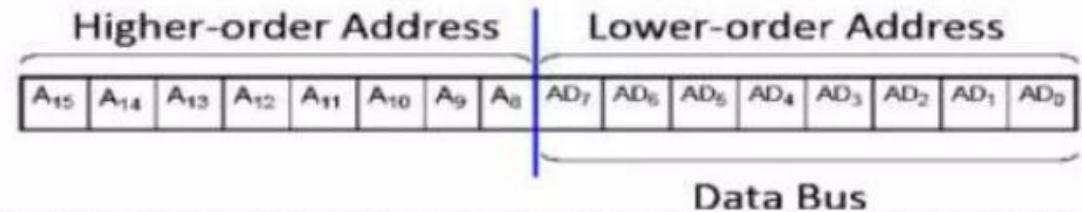
✓ CLK (output): (**Pin 37**)

- Clock Output is used as the system clock for peripheral and devices interfaced with the microprocessor.

Buses: Address and Data

Address Bus (Pin 21-28)

- 16 bit address lines A₀ to A₁₅
- The address bus has 8 signal lines A₈ – A₁₅ which are unidirectional.
- The other 8 address lines A₀ to A₇ are multiplexed (time shared) with the 8 data bits.



Data Bus (Pin 19-12)

- To reduce the number of pins lower order address pin are multiplexed with 8 bit data bus (bidirectional)
- So, the bits AD₀ – AD₇ are bi-directional and serve as A₀ – A₇ and D₀ – D₇ at the same time.
- During the execution of the instruction, these lines carry the address bits during the early part (T₁ state), then during the late parts(T₂ state) of the execution, they carry the 8 data bits.

Control Signals: 4

Control Pins – RD, WR, IO/M(active low)

1. RD: Read(Active low) (Pin 32)

- Read Memory or I/O device
- Indicated that data is to be read either from memory or I/P device and data bus is ready for accepting data from the memory or I/O device.

2. WR: Write(Active low) (Pin 31)

- Write Memory or I/O device
- Indicated that data on the data bus are to be written into selected memory or I/P device.

3. IO/M: (Input Output/Memory-Active low) (Pin 34)

- Signal specifies that the read/write operation relates to whether memory or I/O device.
- When (IO/M=1) the address on the address bus is for I/O device
- When (IO/M=0) the address on the address bus is for memory

IO/M(active low)	RD	WR	Control Signal	Operation
0	0	1	MEMR	M/M Read
0	1	0	MEMW	M/M write
1	0	1	IOR	I/O Read
1	1	0	IOW	I/O Write

Status Configuration

When S_0, S_1 is combined with IO/M(active low), we get status of machine cycle

IO/M	S_1	S_0	OPERATION	Control Signal
0	1	1	Opcode fetch	$\overline{RD} = 0$
0	1	0	Memory read	$\overline{RD} = 0$
0	0	1	Memory write	$\overline{WR} = 0$
1	1	0	I/O read	$\overline{RD} = 0$
1	0	1	I/O write	$\overline{WR} = 0$
1	1	0	Interrupt Acknowledge	$\overline{INTA} = 0$
Z	0	0	Halt	$\overline{RD}, \overline{WR} = Z$ and $\overline{INTA} = 1$
Z	X	X	Hold	
Z	X	X	Reset	

Z= Tristate, X = don't care condition

DMA for 8085

DMA:

- When 2 or more devices are connected to a common bus, to prevent the devices from interfering with each other, the tristate gates are used to disconnect all devices except the one that is communicating at a given instant .
- The CPU controls the data transfer operation between memory and I/O device.
- DMA operation is used for large volume data transfer between memory and an I/O device directly.
- The CPU is disabled by tri-stating its buses and the transfer is effected directly by external control circuits.

✓ **HOLD (Pin 38)**

- This signal indicates that another device is requesting the use of address and data bus.
- So it relinquish the use of buses as soon as the current machine cycle is completed.
- MP regains the bus after the removal of a HOLD signal

✓ **HLDA (Pin 39)**

- On receipt of HOLD signal, the MP acknowledges the request by sending out HLDA signal and leaves out the control of the buses.
- After the HLDA signal the DMA controller starts the direct transfer of data.
- After the removal of HOLD request HLDA goes low.

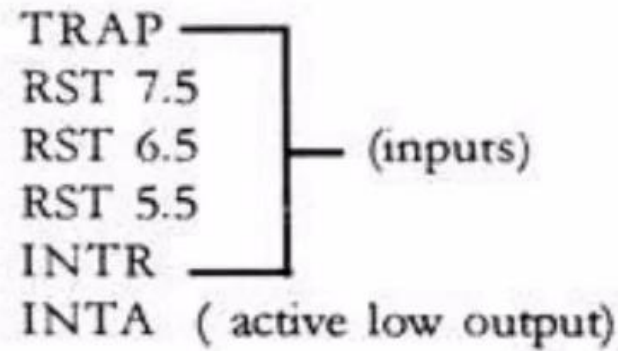


Ready (input) (Pin 35)

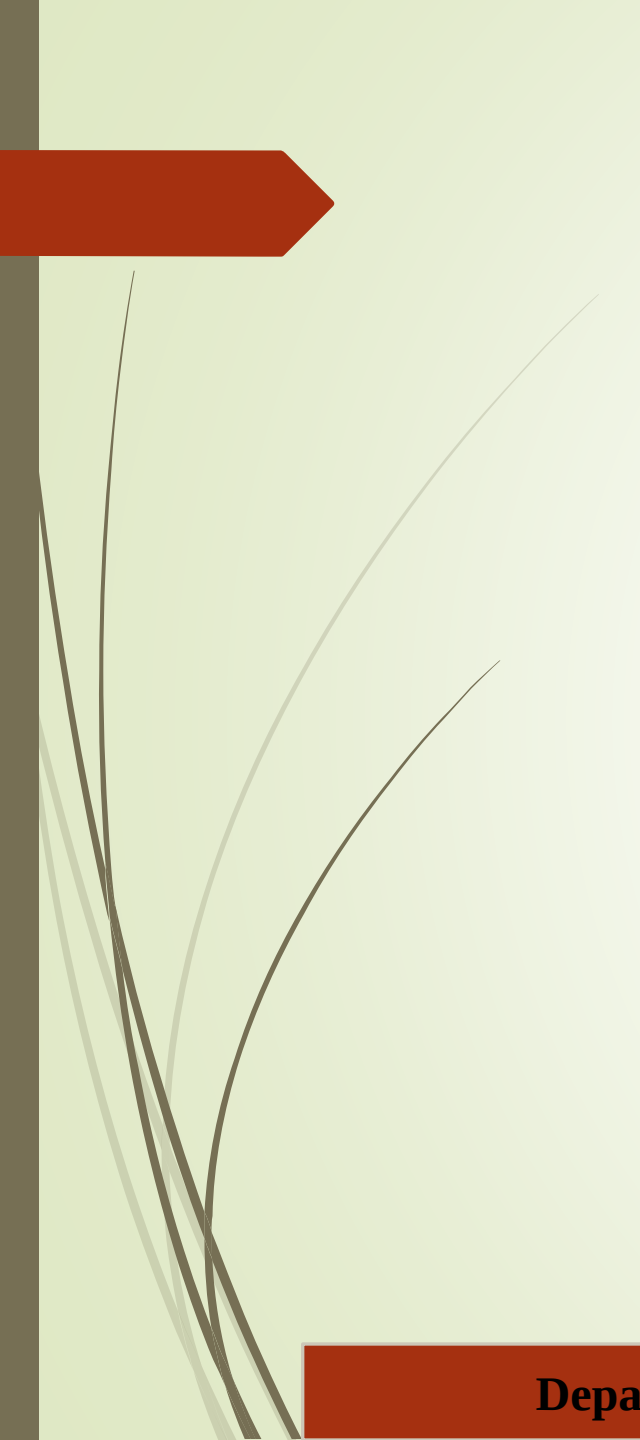
- Memory and I/O devices will have slower response compared to microprocessors.
- Before completing the present job, such a slow peripheral may not be able to handle further data or control signal from CPU.
- The processor sets the READY signal after completing the present job to access the data.
- It synchronizes slower peripheral to the processor.
- The microprocessor enters into WAIT state while the READY pin is disabled.

Interrupt Signals

- They are the signals initiated by an external device to request the microprocessor to do a particular task or work.
- There are five hardware interrupts called, **(Pin 6-11)**



- On receipt of an interrupt, the microprocessor acknowledges the interrupt by the active low INTA (Interrupt Acknowledge) signal.



Thanks